QUESTION BANK – BMATE24401

MODULE – 1

1. The probability density function of a variable X is given by the following table

X	0	1	2	3	4	5	6
$P(X)$	K	$3K$	$5K$	$7K$	$9K$	$11K$	$13K$

For what value of K this represents a valid probability distribution also find $P(X \geq 5), P(3 < X \leq 6), P(X < 4)$

2. Find the value of k, mean and variance for the following distribution:

X	-2	-1	0	1	2	3
$P(X)$	0.1	K	0.2	$2K$	0.3	K

Also find $P(X \geq 2), P(X < 2), P(X > -1), P(-1 < X \leq 2), P(-2 < X < 2)$

3. The probability distribution of a finite random variable X is given by the following table:

X	0	1	2	3	4	5	6	7
$P(X)$	0	K	$2K$	$2K$	$3K$	K^2	$2K^2$	$7K^2 + K$

Find $K, P(X < 6), P(X \geq 6), P(3 < X \leq 6)$

4. A coin is tossed 4 times. Find the probabilities of getting (i) Exactly 1 head, (ii) At most 3 heads, (iii) At least 2 heads

5. The probability that the bomb dropped from a plane will strike the target is $\frac{1}{5}$. If 6 bombs are dropped find the probability that (i)exactly 2 will strike the target, (ii) atleast 2 will strike the target.

6. A random variable X has the density function $f(x) = \begin{cases} Kx^2, -3 < x < 3 \\ 0, otherwise \end{cases}$
Find $K, P(1 \leq X \leq 2), P(X \leq 2), P(X > 1)$,

7. Out of 800 families with 4 children each, how many families would be expected to have (i)2 boys and 2 girls, (ii) At least 1 boy, (iii) At most 2 girls. Assume equal probabilities for boys and girls. Corresponding each child as a trial $n = 4$. Assume that birth of a boy is a success $p = \frac{1}{2}, q = \frac{1}{2}$

8. The probability that a man aged 60 will live upto 70 is 0.65. Out of 10 men now aged 60, find the probability that (i)at least 7 will live upto 70, (ii)Exactly 9 will live upto 70, (iii) Atmost 9 will live upto 70

9. If the chance that one of the 10 telephone lines is busy at an instant is 0.2.
(i) What is the chance that 5 of the lines are busy
(ii) What is the probability that all the lines are busy
(iii)What is the probability that atleast one line is busy

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10. The number of accidents in a year to taxi drivers in a city has mean 3 out of 1000 taxi drivers. Find approximately how many can be expected to have (i) No accident in a year, (ii) More than 3 accidents in a year, (iii) Less than 5 accidents in a year.

11. A certain screw making machine produces on an average two defectives out of 100 and packs them in boxes of 500. Find the probability that the box contains (i) Three defectives, (ii) Atleast one defective, (iii) Between 2 and 4 defectives

12. In a certain factory turning out blades there is a small chance of 0.002 for any blade to be defective. The blades are supplied in a packets of 10. Using Poisson distribution find the approximate number of packets containing (i) No defective blade, (ii) One defective blade in a consignment of 10000 packets.

13. In a certain town the duration of shower is exponentially distributed with mean 5 minute. What is the probability that a shower will last (i) 10 minutes or more, (ii) less than 10 minutes, (iii) Between 10 and 12 minutes.

14. The length of telephone conversation is an exponential variate with mean 3 minutes. Find the probability that a call (i) ends in less than 3 minutes, (ii) takes between 3 to 5 minutes.

15. In a test on 2000 electric bulbs, it was found that the life of a particular make was normally distributed with an average life of 2040 hours and S.D. of 60 hours. Estimate the number of bulbs likely to burn for (i) more than 2150 hours, (ii) less than 1950 hours, (iii) more than 1920 hours and less than 2160 hours. Given $P(0 < Z < 1.83) = 0.4664, P(0 < Z < 1.5) = 0.4332, P(0 < Z < 2) = 0.4772$

16. In a normal distribution, 31% of the items are under 45 and 8% are over 64. Find the mean and standard deviation given that $A(0.5) = 0.19, A(1.5) = 0.42$. Where $A(t)$ is the area under the standard normal curve from 0 to t.

17. 100 students appeared in an examination, the distribution of marks is assumed to be normal with mean $= \mu = 70$ and $S.D. = \sigma = 5$, how many students are expected to get marks (i)less than 65 marks. (ii) more than 75 (iii) between 65 and 75 marks.Given $A(1) = 0.3413$

18. Suppose X and Y are independent random variables with the following distributions. Find the joint distribution X and Y, Verify that $COV(X, Y) = 0$ given

X	1	2
$f(X)$	0.7	0.3

Y	-2	5	8
$g(Y)$	0.3	0.5	0.2

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19. The joint distribution of two random variables X and Y is given. In each case, find (i) the distributions of X and Y, (ii) $COV(X,Y)$ (iii) Are X and Y independent?

$X\backslash Y$	-4	2	7
1	1/8	1/4	1/8
5	1/4	1/8	1/8

20. The joint probability distributions of $X$and $Y$is given below. Determine (a) marginal distributions of X and Y (b) $Cov(X,Y)$ (c) $\rho(X,Y)$

$X\backslash Y$	-2	-1	4	5
1	0.1	0.2	0	0.3
2	0.2	0.1	0.1	0

21. Given that 2% of the fuses manufactured by a firm are defective. Find the probability that a box containing 200 fuses has (i)Atleast 1 defective fuse, (ii) 3 or more defective fuses, (iii) No defective fuse.

22. If the probability of bad reaction from certain injection is 0.001. Determine the chance that out of 2000 individuals, more than 2 will get bad reaction.

23. The sales per day in a shop is exponentially distributed with average sale amounting to Rs. 100 and net profit is 8%. Find the probability that the net profit exceeds (i)Rs. 30 on a day, (ii) Rs. 30 on two consecutive days

24. A fair coin is tossed 3 times. Let Xdenote the random variable 0 or 1 accordingly as a head or a tail occurs on the first toss and let Y be the random variable representing the total number of heads that occurs. Find the (i) joint distribution function of (X,Y).(ii) marginal distributions of X and Y (iii) Expectations of X, Y and XY (iv) Covariance of X and Y.

25. The joint probability function of two discrete random variables X and Y is given by

$$f(x) = \begin{cases} c(2x+y), 0 \le x \le 2, 0 \le y \le 3 \\ 0, otherwise \end{cases}$$ Find $i)$ constant c

$ii)$ $P[X=1, Y=3]$ $iii)$ $P[X \ge 1, Y \le 2]$ $iv)$ Marginal probability distributions of X and Y

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MODULE – 2

1. The mean life time of a sample of 100 fluorescent light bulbs produced by a company is computed to be 1570 hours with standard deviation of 120 hours. If μ is the life time of all the bulbs produced by the company test the hypothesis $\mu = 1600$ hours against the alternative hypothesis $\mu \neq 1600$ hours using a level of significance of 0.01.Given $z_\alpha = 2.58$

2. A coin was tossed 400 times and the head turned up 216 times. Test the hypothesis at 5% level of significance that coin is unbiased. Given $z_\alpha = 1.96$

3. A die was thrown 9000 times and a throw of 5 or 6 was obtained 3240 times on the assumption of random throwing, do the data indicate that the die is unbiased?.Given $z_\alpha = 1.96$

4. A random sample of 100 recorded deaths in past year showed an average life span of 71.8 years. Assuming a population standard deviation of 8.9 years, does the data indicate that the average span today is greater than 70 years ? Use 0.05 level of significance. Given $z_\alpha = 1.645$

5. The standard deviation of marks scored by candidates in CET is $\sigma = 10.5$. The mean marks scored by 64 randomly selected candidates is 38. Test at 1% level of significance whether the average CET marks is less than 40?.Given $z_\alpha = -2.33$

6. The mean household income in a region served by a chain of clothing stores is Rs. 38, 750. In a sample of 50 customers taken at various stores the mean income of the customers was Rs. 41, 505 with standard deviation Rs. 7, 652. Test at the 1% level of significance the null hypothesis that the mean household income of customers of the chain is Rs. 38, 750 against that alternative that it is different from Rs. 38, 750.Gven $z_\alpha = 2.58$

7. A sample of 100 bulbs produced by a company showed a mean life of 1190 hours and S.D of 90 hours. Also a sample of 75 bulbs produced by a company B showed a mean life of 1230 hours and a S.D of 120 hours. Is there a difference between the mean life time of the bulbs produced by the two companies at 1% level of significance.Given $z_\alpha = 2.58$

8. A die was thrown 1200 times and the number 6 was obtained 236 times. Can the die is considered as fair at 0.01 level of significance. Given$z_\alpha = 2.58$

9. A mechanist makes engine parts with axle diameter of 0.7 inch. A random sample of 10 parts shows mean diameter of 0.742 inch with SD of 0.04 inch. On the basis of this sample, would you say that the work is inferior at 5% level of significance given $t_{0.05}(9) = 2.262$

10. Ten oil tins are taken at random from an automatic filling machine. The mean weight of the tins 15.8kg and standard deviation of 0.5 kg. Does the sample mean differ significantly from the intended weight of 16 kg?. Use $t_{0.05}$ (9) = 2.26

11. A random sample of 10 boys had the following I.Q's: 70, 120, 110, 101, 88, 83, 95, 98, 107, and 100. Do these data support the assumption of a population mean I.Q of 100? Find the reasonable range in which most of the mean I.Q values of samples of 10 boys lie. Use $t_{0.05}$ (9) = 2.26

12. Two horses A and B were tested according to the time (in secs) to run a particular race with the following results

Horse A	28	30	32	33	33	29	34
Horse B	29	30	30	24	27	29	

Test whether you can discriminate between two horses Use $t_{0.05}$ (11) = 2.2

13. Fit a poisson distribution to the following data and test the goodness of fit at 5% level of significance. Given $\chi^2_{0.05}(3) = 3.82$

x	0	1	2	3	4
f	419	352	154	56	19

14. Ten individuals are chosen at random from a population and their heights in inches are found to be $63, 63, 64, 67, 68, 69, 70, 70, 71, 71$. Test the hypothesis that the mean of the universe is 66 inches use $t_{0.05} = 2.262$ for 9 df.

15. A certain stimulus administered to each of the 12 patients resulted in the following change in blood pressure $5, 2, 8, -1, 3, 0, 6, -2, 1, 5, 0, 4$. Can it be concluded that the stimulus will change the blood pressure? Use $t_{0.05}(11) = 2.201$.

16. A group of 10 boys fed on a diet A and another group of 8 boys fed on a diet B for a period of 6 Months recorded the following increase in weights

Diet A	5	6	8	1	12	4	3	9	6	10
Diet B	2	3	6	8	10	1	2	8		

Whether diets A and B differ significantly regarding their effects on increase in weights. Use $t_{0.05}(10) = 2.12$

17. Five dice were thrown 96 times and the number 1, 2 or 3 appearing on the face of the dice follows the frequency distribution as follows:

No. of dice showing 1 or 2 or 3	5	4	3	2	1	0
Frequency	7	19	35	24	8	3

Test the hypothesis that the data follows a Binomial distribution. Given $\chi^2_{0.05}(5) = 11.07$

18. A set of 5 similar coins are tossed 320 times and the observations are

No. of heads	0	1	2	3	4	5
Frequency	6	27	72	112	71	32

Test the hypothesis that the data follows a Binomial distribution. For 5 df we have $\chi^2_{0.05}(5) = 11.07$

19. A dice is thrown 264 times and the number appearing on the face (x) follows the following frequency distribution. Show that the die is biased $\chi^2_{0.05}(5) = 11.07$

x	0	1	2	3	4	5
f	40	32	28	58	54	60

20. Fit a Poisson distribution for the following data and test goodness of fit given the values of chi-square for 3 df at 5% level of significance is 7.815.

x	0	1	2	3	4
f	122	60	15	2	1

21. Fit a Poisson distribution for the following data and test goodness of fit given the values of chi-square for 3 df at 5% level of significance is 7.815.

x	0	1	2	3	4
f	90	70	30	8	2

22. Five dice were thrown 96 times and the number 1, 2 or 3 appearing on the face of the dice follows the frequency distribution as follows:
Test the hypothesis that the data follows a Binomial distribution. $\chi^2_{0.05}(5) = 11.07$

No. of dice showing 1 or 2 or 3	5	4	3	2	1	0
Frequency	10	22	38	30	15	5

23. A set of 5 similar coins are tossed 400 times and the observations are as follows. Test the hypothesis that the data follows a Binomial distribution. For 5 df we have $\chi^2_{0.05}(5) = 11.07$

No. of heads	0	1	2	3	4	5
Frequency	5	30	90	140	95	40

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MODULE - 3

1. Fit a parabola of second degree $y = a + bx + cx^2$ for the data

X	0	1	2	3	4
Y	1	1.8	1.3	2.5	2.3

2. If P is a pull required to lift a load W by means of a pulley block, find a linear law of the form $P = mW + c$ connecting P and W using the following data. Where P and W are taken in kg-wt compute P when $W = 150kg$

P	12	15	21	25
W	50	70	100	120

3. A data science team is investigating how the size of a training dataset influences the prediction accuracy of a machine learning model. The following observations were recorded during experimentation:

x	1	2	3	4	6	8
y	2.4	3.0	3.6	4.0	5.0	6.0

Assuming a linear relationship between dataset size and accuracy, determine the equation of the best-fit line $y = ax + b$ using the method of least squares.

4. In a data analytics experiment, a computer system records the relationship between the size of input data x (in GB) and the corresponding processing time y (in seconds). Due to system overhead and resource scaling,the relationship is observed to be nonlinear. The collected experimental data are given below.

x	-2	-1	0	1	2
y	-3.150	-1.390	0.620	2.880	5.378

Assuming that the relationship between processing time and input size follows a quadratic model $y = a + bx + cx^2$ Use the method of least squares to determine the best-fit values of the parameters a, b and c

5. A traffic engineer is studying the effect of traffic signal cycle length on the average vehicle delay at a busy urban intersection. Due to varying traffic flow conditions, the relationship between signal timing and delay is expected to be nonlinear. The following observations were recorded during field measurements:

x	2	4	6	8	10
y	3.07	12.85	31.47	57.38	91.29

Assuming that the relationship between signal cycle length and the average delay can be modeled by a quadratic equation, fit a second-degree parabola $y = a + bx + cx^2$ using the method of least squares. Determine the values of the parameters a, b and c

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6. Fit a second-degree parabola $y = a + bx + cx^2$ in the least square sense for the following data and hence estimate y at $x = 6$

x	1	2	3	4	5
y	10	12	13	16	19

7. Fit a straight line $y = ax + b$ for the following data by method of least squares

x	1	3	4	6	8	9	11	14
y	1	2	4	4	5	7	8	9

8. During the evaluation of a computational system, an analyst studies the relationship between a control parameter (X) and the system performance metric (Y). Because of randomness in input data, hardware variability, and algorithmic effects, the observed measurements show fluctuations and are recorded as,

X	1	2	3	4	5	6
Y	6	4	3	5	4	2

Using the method of least squares, determine the best-fit straight line that models the relationship between the control parameter and the performance metric. Use the model to predict the value of Y when X = 4.

9. Find the coefficient of correlation and lines of regression using the following data

x	1	2	3	4	5	6	7	8	9	10
y	10	12	16	28	25	36	41	49	40	50

10. Obtain the lines of regression and hence find the coefficient of correlation of the data

x	1	2	3	4	5	6	7
y	9	8	10	12	11	13	14

11. From the following data find the coefficient of correlation between industrial production and exports (in crore tone)

x	55	56	58	59	60	61	62
y	35	38	38	39	44	43	45

12. Calculate the rank correlation coefficient from the following data

x	56	42	72	36	63	47	55	49	38	41	68	60
y	147	125	160	118	149	128	150	145	115	140	152	155

13. The following data gives the age of husband (x) and the age of wife (y) in years. Form the two regression lines and calculate the age of husband corresponding to the 16 years age of wife.

x	36	23	27	28	28	29	30	31	33	35
y	29	18	20	22	27	21	29	27	29	28

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14. Given the equation of the regression lines $x = 19.13 - 0.87y$, $y = 11.67 - 0.5x$. Compute the means of x and y and the coefficient of correlation.

15. The regression equations of two variables x and y are $3x + 2y = 26$ and $6x + y = 31$, find means of x and y and correlation coefficient r.

16. In a partially destroyed lab record on correlation data, the following results are only available, var(x)=9, regression equations are $4x - 5y + 33 = 0$ and $20x - 9y = 107$. Find mean of x, y, and standard deviations of x and y

17. Calculate the rank correlation coefficient from the following data

Marks in Physics	36	43	47	28	35	50	40
Marks in Mathematics	73	44	35	30	20	36	40

18. Calculate the rank correlation coefficient from the following data

x	68	64	75	50	69	80	76	40	55
y	62	58	68	45	81	60	69	48	50

19. Compute the means of x and y and find the coefficient of correlation r from the given regression lines $2x + 3y + 1 = 0$ and $x + 6y - 4 = 0$

20. In a beauty contest 10 participants are ranked by 3 judges. Determine which pair of judges has a common taste with respect to beauty

Judge 1	4	7	2	1	5	10	9	6	3	8
Judge 2	6	5	1	2	9	10	3	7	4	8
Judge 3	6	8	2	1	5	9	10	3	4	7

21. The rank of 10 students of some batch in two subjects A and B are given below. Calculate the rank correlation coefficient.

Rank A	1	2	3	4	5	6	7	8	9	10
Rank B	6	7	5	10	3	9	4	1	8	2

22. Calculate the rank correlation coefficient for:

x	12	18	25	30	45	50	60	72
y	15	20	28	35	40	55	65	75

23. Find Spearman's rank correlation coefficient

x	5	10	20	25	30	35	38
y	6	9	14	22	24	33	30

24. Compute the rank correlation coefficient:

x	68	75	80	60	55	90	85
y	70	78	85	60	88	82	65

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MODULE - 4

1. Find the Analytic function f(z) whose imaginary part is $e^x\,(x\sin y + y\cos y)$
2. Construct the analytic function whose imaginary part is $-\frac{\sin\theta}{r}$
3. Determine the analytic function whose real part is $y + e^x cosy$
4. Find the analytic function $f(z) = u + iv$ given that $u = x^2 - y^2 + \frac{x}{x^2+y^2}$
5. Show that the $f(z) = z + e^z$ is analytic and find its derivative.
6. Show that $f(z) = z^n$ is analytic and find its derivative.
7. Show that $f(z) = log(z)$ is analytic and find its derivative.
8. Show that $v = 2xy - 2x + 4y$ is harmonic and find its harmonic conjugate and also find f(z).
9. Show that $u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$ is harmonic and find its harmonic conjugate also find f(z).
10. Show that the function $u = sinxcoshy + 2cosxsinhy + x^2 - y^2 + 4xy$is a harmonic function and determine the corresponding analytic function
11. Show that $u = (r - \frac{1}{r})cos\theta$ is harmonic, find its harmonic conjugate and also find its analytic function
12. Given $u - v = (x - y)(x^2 + 4xy + y^2)$ find the analytic function $f(z) = u + iv$
13. In an Electromagnetic field XY plane is given by the potential function $\phi = x^2 - y^2$ find the stream function also find the complex potential.
14. Show that $u = e^{2x}(xcos2y - ysin2y)$ is harmonic and hence find the analytic function.
15. Show that $u = \frac{1}{r}cos\theta$ is harmonic and find its harmonic conjugate and also find its analytic function.
16. If f(z) is a regular function show that $\left(\frac{\partial}{\partial x}|f(z)|\right)^2 + \left(\frac{\partial}{\partial y}|f(z)|\right)^2 = |f'(z)|^2$
17. Find the analytic function $f(z) = u + iv$ given that $u - v = e^x(cosy - siny)$.
18. In a 2-dimensional fluid flow the stream function $\psi = \tan^{-1}(\frac{y}{x})$, find potential function ϕ and f(z)
19. In an Electromagnetic field XY plane is given by potential function $\phi = 3x^2y - y^3$ find stream function, also find complex potential.
20. If f(z) is a regular function show that $\left\{\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right\}|f(z)|^2 = 4|f'(z)|^2$
21. Find the analytic function $f(z) = u + iv$ given,$v = (r - \frac{1}{r})sin\theta, r \neq 0$

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MODULE – 5

1. Discuss the transformation $w = e^z$

2. Find bilinear transformation that maps $0, -i, -1$ of $z - plane$ onto the points $i, 1, 0$ of $w - plane$ respectively.

3. Find the bilinear transformation which maps the points $1, i, -1$ of $z - plane$ onto the points $2, i, -2$ of $w - plane$ respectively.

4. Find the bilinear transformation that maps the points $z = -1, i, 1$onto the points $w = 1, i, -1$ respectively.

5. Find the Bilinear transformation that maps the points $z = 1, i, -1$ respectively onto the points $w = i, 0, -i$

6. Discuss the transformation $w = z^2$

7. Find bilinear transformation which maps the points $z = 1, i, -1$onto the points $w = 0, 1, \infty$ respectively.

8. Find bilinear transformation which maps the points $z = 0, 1, \infty$ onto the points $w = -5, -1, 3$ respectively.

9. Evaluate $\int_0^{1+i}(x^2 + iy)\, dz$ along the paths: i) $y = x$ and ii) .$y = x^2$

10. Evaluate $\int_c z^2\, dz$
    a) Along the straight line from $z = 0$, to $z = 3 + i$
    b) Along the curve made up of two line segments, one from $z = 0$ to $z = 3$ and another from $z = 3$ to $z = 3 + i$

11. Evaluate $\int_0^{2+i} z^2\, dz$ along
    i) The line $y = \frac{x}{2}$ or $x = 2y$
    ii) The real axis to 2 and then vertically to $2 + i$.

12. Evaluate $\int_c \frac{z^2-z+1}{z-1}\, dz$ where c is the circle
    i) $|z| = 1$ ii) $|z| = \frac{1}{2}$

13. Evaluate $\int_c \frac{\sin\pi z^2+\cos\pi z^2}{(z-1)(z-2)}\, dz$ where $c : |z| = 3$

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14. Evaluate $\int_c |z|^2 dz$ where C is a square with the following vertices: (0, 0), (1, 0), (1, 1) and (0, 1).

15. Evaluate $\int_c \frac{z}{(z-1)(z+2)}\, dz$ where C is the circle $|z| = 2$

16. Evaluate $\int_c \frac{\sin^2 z}{(z-\frac{\pi}{6})^3}\, dz$ where $c : |z| = 4$ using Residual theorem.

17. Evaluate $\int_c \frac{e^{2z}}{(z+1)^4}\, dz$ where C is a circle $|z| = 2$ using Residual theorem

18. Evaluate $\int_c \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)^2(z-2)}\, dz$ where $c : |z| = 3$ using Residual theorem

19. Evaluate $\int_c \frac{e^z}{(z+1)^4}\, dz$ where $c : |z| = 3$ using Residual theorem

20. Evaluate $\int_c \frac{e^{2z}}{(z+1)(z+2)}\, dz$ where $c : |z| = 3$ using Cauchy's residue theorem.

21. Evaluate $\int_c \frac{e^{2z}}{(z+1)^2(z-2)}\, dz$ where $c : |z| = 3$ using Residual theorem.

22. Evaluate $\int_c \frac{z}{z^2-3z+2}\, dz$ where $c : |z-2| = \frac{1}{2}$

23. Evaluate $\int_c \frac{dz}{(z^2-4)}$ where (i) $c : |z+2| = 1$ (ii) $c : |z| = 1$